

CO2-AT 2 – SCENARIO BASED ASSESSMENT

Question 1

Scenario 1: Manufacturing Quality Inspection

Problem: Images of metal components are corrupted by repetitive horizontal stripes due to electrical interference.

1. Type of noise:

- The interference introduces periodic noise (repetitive patterns in frequency domain).

2. Effect on inspection accuracy:

- Periodic stripes obscure fine details.
- Defects such as cracks or scratches may be hidden or misinterpreted.
- Automated detection algorithms fail due to reduced contrast and false patterns.

3. Recommended enhancement technique:

- Frequency domain filtering using Fourier Transform + Notch Filtering.

4. Justification (why frequency domain is better than spatial methods):

- Periodic noise appears as distinct spikes in the frequency spectrum.
- Notch filters can selectively remove these spikes without affecting the rest of the image.
- Spatial filters (like median or Gaussian) blur the image and cannot effectively target periodic interference.

5. Logical sequence:

- Identify periodic noise → Transform to frequency domain → Apply notch filter → Inverse transform → Clean image for defect detection.

Question 2

scenario 2: Autonomous Vehicle Navigation

Problem: Road images during heavy rain contain noise, but lane markings and edges must remain sharp.

1. **Challenge:**

- Noise reduces visibility, but edges (lane boundaries, road edges) are critical for navigation.
- Standard denoising may blur these edges, leading to unsafe driving decisions.

2. **Importance of edge preservation:**

- Lane detection relies on sharp boundaries.
- Blurred edges cause misinterpretation of road structure, risking navigation errors.

3. **Recommended filtering technique:**

- Bilateral Filtering (edge-preserving smoothing).

4. **Justification (why bilateral filter works):**

- Combines spatial proximity and intensity similarity.
- Smooths noise in homogeneous regions (rainy background).
- Preserves sharp intensity changes (lane markings, road edges).
- Unlike Gaussian or median filters, it avoids edge blurring.

5. **Logical sequence:**

- Capture noisy image → Apply bilateral filter → Noise reduced while edges remain intact
→ Reliable Lane detection for safe navigation.